C S Vaishakh

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Professional Summary

Artificial Intelligence & Data Science undergraduate with an interest in full-stack development and machine learning. Experienced in building applications using TypeScript (React, Next.js, Express.js etc.), backend systems with Go (Fiber and Gin, Postgres), AI/ML applications in Python using FastAPI and related machine learning frameworks.

Education

St. Joseph's College of Engineering and Technology, Palai Kottayam, Kerala
B.Tech in Artificial Intelligence and Data Science — APJ Abdul Kalam Technological University *2023 – 2027*

Projects

Quickhand December 2025 – Present

- Implemented role-based access control across two user roles, securing Fiber v3 routes with stateless JWTs.
- Built an authenticated WebSocket layer broadcasting real-time job alerts to matched handymen.
- Structured the codebase as a Go workspace monorepo pairing the Fiber API with a TanStack Start frontend.
- **Technologies**: Go, Fiber v3, WebSockets, React, TanStack Start, TypeScript, PostgreSQL, GORM, Docker

Snip July – August 2026

- Built a URL shortener with a Go (Gin) REST API, React dashboard, JWT auth, and collision-safe codes.
- Cut redirect latency by fronting PostgreSQL with a Redis cache on the public redirect path.
- Deployed the frontend on Vercel and the backend on AWS EC2 behind Caddy via GitHub Actions CI/CD.
- **Technologies**: Go, Gin, GORM, PostgreSQL, Redis, JWT, React, Vite, Docker, AWS EC2, Vercel

PortfolyoIQ March – August 2026

- Engineered a privacy-preserving rebalancing platform: Next.js client, Express API, FastAPI model service.
- Trained logistic regression models in-browser with TensorFlow.js, eliminating raw portfolio data uploads.
- Implemented Federated Averaging (FedAvg) on the Express aggregator to synchronize local model updates.
- Self-hosted all three services on AWS EC2 behind Caddy via systemd, deployed by GitHub Actions.
- **Technologies**: Next.js, Express.js, FastAPI, TensorFlow.js, Drizzle ORM, PostgreSQL, Bun, AWS EC2

ONT Monitor April 2026

- Built a Go microservice (Fiber v3, GORM) ingesting real-time ONT status change events.
- Implemented thread-safe in-memory queues and a worker pool to evaluate outages without race conditions.
- Integrated the Twilio API to dispatch outage and recovery WhatsApp alerts on state transitions.
- **Technologies**: Go, Fiber v3, GORM, PostgreSQL, Twilio API, Docker

Technical Skills

Programming: Python, TypeScript, Go, Java, JavaScript
Web & Backend: Next.js, React, Express.js, Fiber v3, Gin, FastAPI, Node.js, REST APIs, WebSockets
Databases & ORMs: PostgreSQL, MongoDB, Redis, GORM, Drizzle ORM, Mongoose
AI / ML: TensorFlow.js, scikit-learn, NumPy, Pandas, Feature Engineering, Model Evaluation
Tools & DevOps: Git, GitHub Actions (CI/CD), Docker, AWS EC2, Caddy, Vercel, Bun, Linux, Postman